

Full-Signal Comparison of Independent PN-IQ, GMP, and Sparse PNNN

Objective and protocol. Every linear model uses the same 19,661 identification rows (approximately 4% of the capture), its previously frozen DOMP support and regressor matrix, and all 491,520 rows for full-signal evaluation. Ridge uses the fixed value $\lambda = 10^{-5}$ without validation selection or cross-validation. Each column norm is calculated on identification only; coefficients are then returned to the original regressor scale. Full-signal evaluation includes the identification samples and is not an independent generalization test.

Main results. NMSE is $10 \log_{10}(\sum |y - \hat{y}|^2 / \sum |y|^2)$. The PNNN entry is the unchanged frozen network result; Ridge is not applicable to that nonlinear model.

Model	Parameters	NMSE LS (dB)	NMSE Ridge 10^{-5} (dB)	FLOPs/ sample	Additional operations/sample
Independent PN-IQ	358	-38.534	-36.940	1062	12 complex magnitudes, 2 divisions
Independent I/Q without PN	394	-37.810	-36.417	1086	12 complex magnitudes
Parameter-matched GMP	358	-38.170	-36.686	1899	14 complex magnitudes
Sparse PNNN N12	358	-35.900	–	876	14 complex magnitudes, 2 divisions, 12 ELU

Ridge changes only coefficient estimation: it does not change support, parameter count, or inference FLOPs. The sparse-PNNN FLOP count assumes that zero weights are skipped; executing the original full matrices costs 2252 FLOPs/sample. A complex magnitude includes one square root.

Complex-equivalent reference. Separately, the 100-term complex GMP baseline and its coupled PN-IQ real representation remain numerically equivalent for both LS and fixed Ridge. The full-signal Ridge relative prediction error is 7.989×10^{-13} . This statement does not refer to the 179-term parameter-matched GMP in the table.

Quantitative conclusions. Under LS, Independent PN-IQ and parameter-matched GMP both use 358 real parameters; PN-IQ improves NMSE by approximately 0.364 dB and requires 837 fewer FLOPs/sample. The no-PN control uses 394 parameters, 36 more than PN-IQ, yet is approximately 0.723 dB worse under LS. This comparison indicates that the observed gain is not explained by independent I/Q freedom alone.

With the fixed Ridge value, the three linear rows reach -36.940, -36.417, and -36.686 dB, respectively. Ridge therefore does not improve this capture at 10^{-5} , but its regularized estimates preserve every frozen support and inference cost. Sparse PNNN N12 has the lowest analytical count when zero weights are skipped, while PN-IQ has the best NMSE in both displayed linear fits. GMP and the no-PN control avoid the 12 ELU evaluations of the sparse network. Analytical FLOPs do not establish latency, power, or hardware cost.